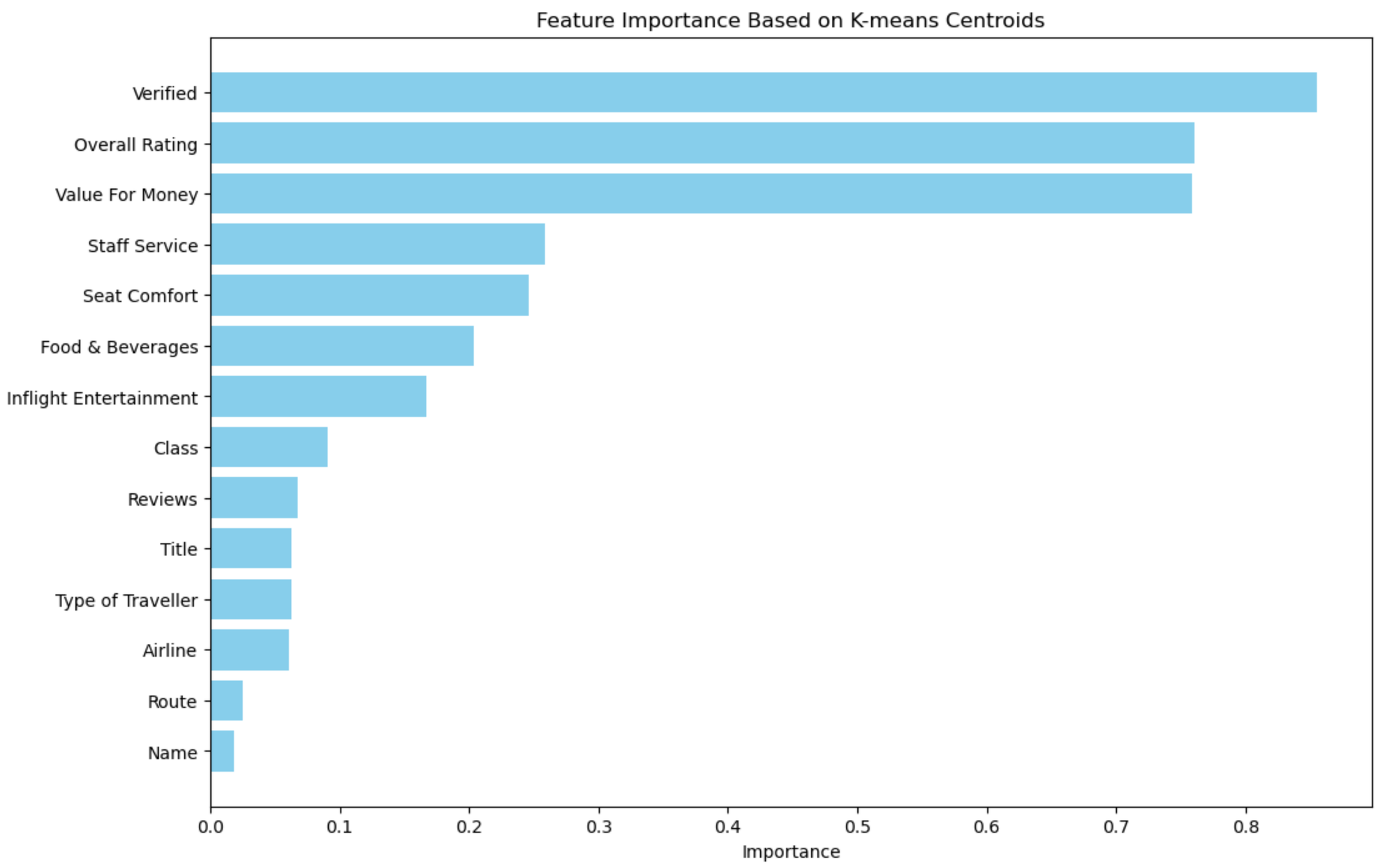
```
>>> import numpy as np
>>> import pandas as pd
>>> from sklearn.model_selection import train_test_split
>>>
>>> # Load the data
>>> data = pd.read_csv('/Users/simonsayin/Documents/Forth Big Data Analytics Certificate/ZERO CODE-MACHINE-Codes-Reviews-Data.xlsx')
>>>
>>> # Display the first few rows of the dataset
>>> print("First 5 rows:")
First 5 rows:
>>> print(data.head())
   Customer ID  Age Gender Purchase Frequency  \
0            1   55   Male                  7
1            2   26 Female                  8
2            3   18   Male                  1
3            4   57   Male                  1
4            5   39 Female                  1

   Purchase Amount (USD) Frequency of Purchases  \
0                     23                    NaN
1                     62                    NaN
2                     73                    NaN
3                     90                    NaN
4                     49                    NaN

   Feedback Score  Loyalty Score  Purchase Frequency Category
0             NaN            NaN                          NaN
1             NaN            NaN                          NaN
2             NaN            NaN                          NaN
3             NaN            NaN                          NaN
4             NaN            NaN                          NaN
>>> print("\n")

>>> # Check for missing values
>>> print("Missing values:")
Missing values:
>>> print(data.isnull().sum())
Customer ID                         0
Age                                 0
Gender                              0
Purchase Frequency                  0
Purchase Amount (USD)               0
Frequency of Purchases            100
Feedback Score                    100
Loyalty Score                     100
Purchase Frequency Category       100
dtype: int64
>>> print("\n")

>>> # Check for duplicate rows
>>> print("Duplicate rows:")
Duplicate rows:
>>> print(data.duplicated().sum())
0
>>> print("\n")

>>> # Check the data after conversion
>>> print("Data after conversion to binary:")
Data after conversion to binary:
>>> print(data.head())
   Customer ID  Age Gender Purchase Frequency  \
0            1   55   Male                  7
1            2   26 Female                  8
2            3   18   Male                  1
3            4   57   Male                  1
4            5   39 Female                  1

   Purchase Amount (USD) Frequency of Purchases  \
0                     23                    NaN
1                     62                    NaN
2                     73                    NaN
3                     90                    NaN
4                     49                    NaN

   Feedback Score  Loyalty Score  Purchase Frequency Category
0             NaN            NaN                          NaN
1             NaN            NaN                          NaN
2             NaN            NaN                          NaN
3             NaN            NaN                          NaN
4             NaN            NaN                          NaN
>>> print("\n")

>>> # Check columns
>>> print("Columns:")
Columns:
>>> print(data.columns)
Index(['Customer ID', 'Age', 'Gender', 'Purchase Frequency',
       'Purchase Amount (USD)', 'Frequency of Purchases',
       'Feedback Score', 'Loyalty Score',
       'Purchase Frequency Category'],
      dtype='object')
>>> print("\n")

>>> # Check data types
>>> print("Data types:")
Data types:
>>> print(data.dtypes)
Customer ID                         int64
Age                                 int64
Gender                             object
Purchase Frequency                 object
Purchase Amount (USD)              object
Frequency of Purchases             object
Feedback Score                     object
Loyalty Score                      object
Purchase Frequency Category        object
dtype: int64
>>> print("\n")

>>> # Check unique values in each column
>>> print("Unique values:")
Unique values:
>>> print(data.nunique())
Customer ID                         5000
Age                                   53
Gender                                 2
Purchase Frequency                     4
Purchase Amount (USD)                 95
Frequency of Purchases                 4
Feedback Score                         5
Loyalty Score                          5
Purchase Frequency Category            3
dtype: int64
>>> print("\n")

>>> # Check data after conversion
>>> print("Data after conversion:")
Data after conversion:
>>> print(data.head())
   Customer ID  Age Gender Purchase Frequency  \
0            1   55   Male                  7
1            2   26 Female                  8
2            3   18   Male                  1
3            4   57   Male                  1
4            5   39 Female                  1

   Purchase Amount (USD) Frequency of Purchases  \
0                     23                    NaN
1                     62                    NaN
2                     73                    NaN
3                     90                    NaN
4                     49                    NaN

   Feedback Score  Loyalty Score  Purchase Frequency Category
0             NaN            NaN                          NaN
1             NaN            NaN                          NaN
2             NaN            NaN                          NaN
3             NaN            NaN                          NaN
4             NaN            NaN                          NaN
>>> print("\n")

>>> # Check columns
>>> print("Columns:")
Columns:
>>> print(data.columns)
Index(['Customer ID', 'Age', 'Gender', 'Purchase Frequency',
       'Purchase Amount (USD)', 'Frequency of Purchases',
       'Feedback Score', 'Loyalty Score',
       'Purchase Frequency Category'],
      dtype='object')
>>> print("\n")

>>> # Check data types
>>> print("Data types:")
Data types:
>>> print(data.dtypes)
Customer ID                         int64
Age                                 int64
Gender                             object
Purchase Frequency                 object
Purchase Amount (USD)              object
Frequency of Purchases             object
Feedback Score                     object
Loyalty Score                      object
Purchase Frequency Category        object
dtype: int64
>>> print("\n")

>>> # Check unique values in each column
>>> print("Unique values:")
Unique values:
>>> print(data.nunique())
Customer ID                         5000
Age                                   53
Gender                                 2
Purchase Frequency                     4
Purchase Amount (USD)                 95
Frequency of Purchases                 4
Feedback Score                         5
Loyalty Score                          5
Purchase Frequency Category            3
dtype: int64
>>> print("\n")

>>> # Check data after conversion
>>> print("Data after conversion:")
Data after conversion:
>>> print(data.head())
   Customer ID  Age Gender Purchase Frequency  \
0            1   55   Male                  7
1            2   26 Female                  8
2            3   18   Male                  1
3            4   57   Male                  1
4            5   39 Female                  1

   Purchase Amount (USD) Frequency of Purchases  \
0                     23                    NaN
1                     62                    NaN
2                     73                    NaN
3                     90                    NaN
4                     49                    NaN

   Feedback Score  Loyalty Score  Purchase Frequency Category
0             NaN            NaN                          NaN
1             NaN            NaN                          NaN
2             NaN            NaN                          NaN
3             NaN            NaN                          NaN
4             NaN            NaN                          NaN
>>> print("\n")

>>> # Check columns
>>> print("Columns:")
Columns:
>>> print(data.columns)
Index(['Customer ID', 'Age', 'Gender', 'Purchase Frequency',
       'Purchase Amount (USD)', 'Frequency of Purchases',
       'Feedback Score', 'Loyalty Score',
       'Purchase Frequency Category'],
      dtype='object')
>>> print("\n")

>>> # Check data types
>>> print("Data types:")
Data types:
>>> print(data.dtypes)
Customer ID                         int64
Age                                 int64
Gender                             object
Purchase Frequency                 object
Purchase Amount (USD)              object
Frequency of Purchases             object
Feedback Score                     object
Loyalty Score                      object
Purchase Frequency Category        object
dtype: int64
>>> print("\n")

>>> # Check unique values in each column
>>> print("Unique values:")
Unique values:
>>> print(data.nunique)
<bmethod: <function DataFrame.nunique at 0x7fb7f7a7a7a0>>
>>> print(data.nunique())
Customer ID                         5000
Age                                   53
Gender                                 2
Purchase Frequency                     4
Purchase Amount (USD)                 95
Frequency of Purchases                 4
Feedback Score                         5
Loyalty Score                          5
Purchase Frequency Category            3
dtype: int64
>>> print("\n")

>>> # Check data after conversion
>>> print("Data after conversion:")
Data after conversion:
>>> print(data.head())
   Customer ID  Age Gender Purchase Frequency  \
0            1   55   Male                  7
1            2   26 Female                  8
2            3   18   Male                  1
3            4   57   Male                  1
4            5   39 Female                  1

   Purchase Amount (USD) Frequency of Purchases  \
0                     23                    NaN
1                     62                    NaN
2                     73                    NaN
3                     90                    NaN
4                     49                    NaN

   Feedback Score  Loyalty Score  Purchase Frequency Category
0             NaN            NaN                          NaN
1             NaN            NaN                          NaN
2             NaN            NaN                          NaN
3             NaN            NaN                          NaN
4             NaN            NaN                          NaN
>>> print("\n")

>>> # Check columns
>>> print("Columns:")
Columns:
>>> print(data.columns)
Index(['Customer ID', 'Age', 'Gender', 'Purchase Frequency',
       'Purchase Amount (USD)', 'Frequency of Purchases',
       'Feedback Score', 'Loyalty Score',
       'Purchase Frequency Category'],
      dtype='object')
>>> print("\n")

>>> # Check data types
>>> print("Data types:")
Data types:
>>> print(data.dtypes)
Customer ID                         int64
Age                                 int64
Gender                             object
Purchase Frequency                 object
Purchase Amount (USD)              object
Frequency of Purchases             object
Feedback Score                     object
Loyalty Score                      object
Purchase Frequency Category        object
dtype: int64
>>> print("\n")

>>> # Check unique values in each column
>>> print("Unique values:")
Unique values:
```

```
2     552
Name: count, dtype: int64

Cluster 2 includes the following types of travellers:
Type of Traveller
3    949
1    620
2    506
0    423
Name: count, dtype: int64

Updated data with type of traveller:
   Type of Traveller  Cluster
0     Family Leisure        1
1     Family Leisure        2
2       Solo Leisure        1
4       Solo Leisure        2
7       Solo Leisure        0
8           Business        1
19    Couple Leisure        1
22    Couple Leisure        0
23    Family Leisure        0
25          Business        0
```

In [39]:

```python
import pandas as pd
import numpy as np
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans

# Load the data
airlines_data = pd.read_excel("/Users/yemaquinn/Documents/YorkU Big Data Analytics Certificate/CDSA 1100-042/Airlines-Reviews-Data.xlsx")

# Convert 'Recommended' to binary (1 for 'yes', 0 for 'no')
airlines_data['Recommended'] = airlines_data['Recommended'].map({'yes': 1, 'no': 0})

# Drop rows with NaNs in 'Recommended'
airlines_data = airlines_data.dropna(subset=['Recommended'])

# Separate features and target
X = airlines_data.drop(columns=['Recommended'])
y = airlines_data['Recommended']

# Convert all categorical columns to numeric codes
categorical_columns = X.select_dtypes(include=['object']).columns
for col in categorical_columns:
    X[col] = pd.Categorical(X[col]).codes

# Select numeric columns
numeric_columns = X.select_dtypes(include=[np.number]).columns
X_numeric = X[numeric_columns]

# Handle missing values for numeric columns
imputer = SimpleImputer(strategy='mean')
X_numeric = imputer.fit_transform(X_numeric)

# Standardize the features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X_numeric)

# Fit KMeans
kmeans = KMeans(n_clusters=3, random_state=42)
clusters = kmeans.fit_predict(X_scaled)

# Add the cluster labels to the original DataFrame
airlines_data['Cluster'] = clusters

# Get cluster centroids
centroids = kmeans.cluster_centers_

# Compute feature importance based on centroids
feature_importance = np.abs(centroids).mean(axis=0)
feature_importance_df = pd.DataFrame({
    'Feature': numeric_columns,
    'Importance': feature_importance
}).sort_values(by='Importance', ascending=False)

# Display feature importance
print("\nFeature importance based on K-means centroids:")
print(feature_importance_df)

# Additional check for feature names and their importance
import matplotlib.pyplot as plt

plt.figure(figsize=(12, 8))
plt.barh(feature_importance_df['Feature'], feature_importance_df['Importance'], color='skyblue')
plt.xlabel('Importance')
plt.title('Feature Importance Based on K-means Centroids')
plt.gca().invert_yaxis()
plt.show()
```

```
Feature importance based on K-means centroids:
                   Feature  Importance
3                 Verified    0.854782
13          Overall Rating    0.760162
12         Value For Money    0.758773
9            Staff Service    0.258583
8             Seat Comfort    0.246180
10        Food & Beverages    0.203475
11  Inflight Entertainment    0.167424
7                    Class    0.090842
4                  Reviews    0.067403
0                    Title    0.062912
5        Type of Traveller    0.062361
2                  Airline    0.060503
6                    Route    0.025370
1                     Name    0.018721
```



In [ ]: